

STORING HEAT ENERGY

TEMPERATURE

The most obvious effect of heat energy on human is the control of our body temperature. We wear extra clothes and sweaters during the harshest cold to insulate the heat generated internally from our body and thus help to keep us warm. We ~~for~~ ourselves to take off excess heat to cool our body temperature. Both of these processes involve transformations of energies.

When the weather is cold, part of the food we eat is converted to chemical energy of our body. The regulatory mechanism of our body tends to keep our body temperature constant during harshest time, we wear extra clothes and sweaters to insulate us from excess cold. Most people tend to eat more during harshest time in order to generate more heat, when and we may also sleep longer to conserve our energy. When the weather is hot, we tend to eat less to maintain the same body temperature. Excess heat energy flows from internal to external body parts. As the ~~for~~ continuously cools the external body temperature an overall constant body temperature is maintained.

Several of our means of transport, the car, ~~the~~ car, trains, aeroplane, etc, involve the conversion of chemical energy (the burning of fuel) into heat energy which is subsequently converted into mechanical energy.

the transformation of a chemical energy in the bicycle to energy of motion, some transfer of energy from the rider is involved. We do a certain amount of work when we pedal, we derive our energy from what we eat and we require this energy to do work. When we ride our bicycles over long distance and especially over hills we expend most of our reserved energy, which is transferred to the bicycle through our contact (the pedals etc) which in turn cause the bicycle to move cycling up a hill, we gain kinetic energy and lose potential energy. A certain amount of heat energy is lost through sweat and through the types due to friction.

The study of the interaction between heat and other forms of energy is referred to as thermodynamics.

The study of heat is most commonly described in terms of the rise in temperature of a given object or material to which it is applied. In other words we often describe the heat (thermal) energy in an object by its hotness or coldness. The temperature of an object is therefore a description of the hotness or coldness of the object.

We can develop a more exact concept of temperature by considering what happens when two systems or bodies (e.g. an iron ball and a bowl of water) are placed in thermal contact. The system in thermal contact will change in such a way that its internal energy is redistributed. If these changes cease, we say

Thermal equilibrium is established when the two systems are first brought into thermal contact, heat always flows from the body of higher temperature to that of lower temperature. Temperature is a fundamental quantity like mass, length and time. It is determined by a thermometer whose scale has been calibrated by comparison with a standard, according to certain procedures or operations.

Difference between Heat and Temperature

Heat

1) S.I. unit joule (J)

2) Measured with Calorimeter, measured with thermometer

3) A form of energy

TEMPERATURE

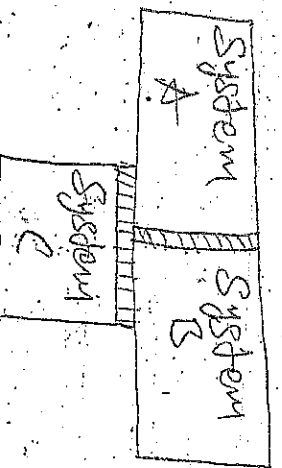
S.I. unit is K or $^{\circ}\text{C}$.

The property that determines the direction of heat flow.

Simultaneous

- 1) They are both scalar quantities.
- 2) Energy is involved in both cases.

Zerth law of thermodynamics states that if a system C is initially in thermal equilibrium with both A and B, then A and B are also in thermal equilibrium with each other.



FTS (43) Zerth law of thermodynamics

Construction and Calibration of mercury glass

Thermometer

One end of a clean capillary tubing is sealed and a bulb is blown there. The open end of the tubing is put beneath the surface of mercury in a trough, and the bulb is gently heated. The air inside expands and escapes through the mercury. On cooling, the air contracts and some mercury runs into the tubing. This is repeated until the bulb and capillary tube are filled. (This should be done ~~carefully~~ to avoid poisoning from mercury vapour). The thermometer is now heated to a temperature a bit higher than the maximum it will measure and the open end is sealed off.

To fix a scale on the thermometer, two fixed temperatures are marked. There are:

- 1) The lower fixed point which is the temperature of pure melting ice and
- 2) The upper fixed point, the temperature of steam from water boiling under standard atmospheric pressure of 760 mm of mercury or 760 mmHg.

- 100°C or 373K - upper fixed point

Thermometer
menometer

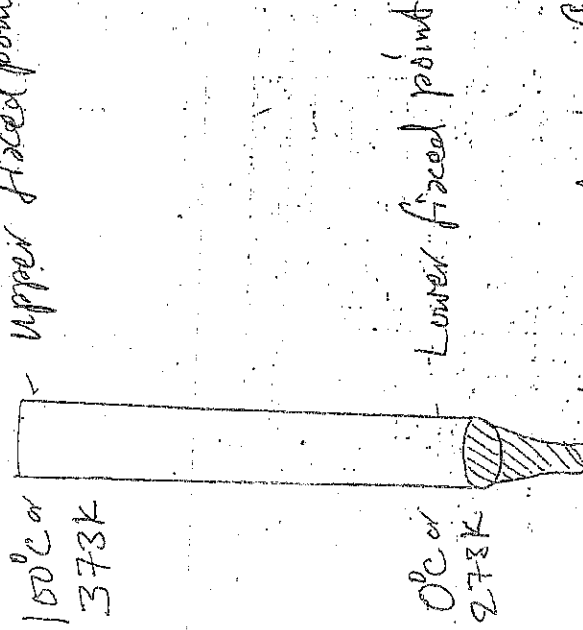


Fig (2) Thermometer: lower and upper fixed point.

Note: Ice must be pure because if there are impurities in it, the melting point will be lowered. Also impurities raise the boiling point of water, so steam is used.

The interval between these two fixed points is divided into a number of equal parts called degree. The lower fixed point 100°C . Another scale used is the SI system, the Kelvin or Absolute scale in which the lower fixed point is 273K and the upper fixed point 373K as shown in Fig (2) above, so $0^{\circ}\text{C} = 273\text{K}$.

Thermometric liquids

Not every liquid that is suitable as thermometric liquid. A liquid used in thermometers must have the following properties.

i) It must have a high boiling point and lower melting point.
ii) It must expand and contract uniformly and readily.

iii) It must be seen easily.

Water does not satisfy any of these properties and hence is unsuitable for use in a thermometer.

The upper fixed point is the temperature of steam from pure water boiling at normal atmospheric pressure.

The lower fixed point is the temperature of a mixture of pure ice and water at normal pressure.

Let compare the merits of mercury and alcohol as thermometric liquid.

Table I

properties	Mercury		Alcohol	
	Advantages	Disadvantages	Advantages	Disadvantages
Freezing point	—	is -39°C , hence not suitable for measuring very low temp ^s . pure.	is -112°C , suitable for measuring very low temperature.	—
Expansion	Regular	Low expansibility	High expansibility, about 10 times that of mercury.	A bit irregular
Boiling point	Very high boiling point 350°C , suitable for measuring high temperature.	—	—	Boils at 78°C , is not suitable for measuring high temperature.
Opacity	Opaque, easily seen	—	—	Not easily seen unless when coloured.
Effect on glass	Does not wet glass	—	—	Wets glass which introduces small reading error.
Evaporation	Not easily evaporated	—	—	Evaporated easily.
Conductivity	High Conductivity and hence responds easily to temperature changes	—	—	Low Conductivity and so has slow response to temperature changes.

Dr. K. S. Chandra

Effects of Heat on Matter

When heat is applied to a body there is,

i) Expansion: When a substance is heated the molecules gain energy and increases which results to an overall increase in the length, size and volume of the substances.

(ii) Rise in temperature: When a substance is heated, the average K.E of its molecules increases, since K.E is directly proportional to its absolute temperature, thus an increase in average K.E results in rise in increases in temperature.

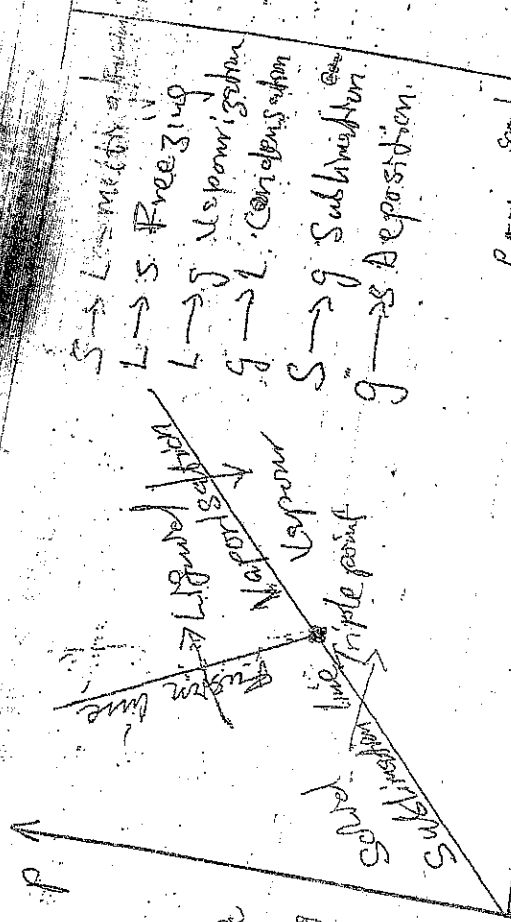
(iii) Change of state: When solids are heated the molecules gain heat energy (K.E) used in breaking down the forces of attraction between its molecules. Molecules no longer hold them in the regular pattern.

As a result, the space - lattice collapse and solid melts. When liquid are heated, the average K.E of its molecules increases. Some molecules are able to acquire or gain enough K.E to be able to break their barrier surfaces to exist as gas vapour.

(iv) Vaporization: It is the process of changing or forming freely the molecules of liquid into bubbles of vapour or gas when its molecules, having acquired maximum K.E are able to break through their

Surface

Triple point: The temperature and pressure at which a substance can exist in equilibrium in the liquid, solid and gaseous is at 0.01°C (273.16 K , 32.01°F) and 4.58 mm (611.2 ps) of mercury and is used to calibrate thermometer.

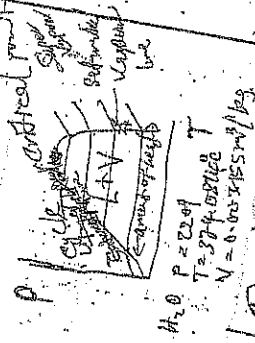


H_2O $0^\circ C$
 $T = 273.15 K$
 $\rho = 0.917 \text{ gm}^3/\text{kg}$

Fig (3) Triple point diagram.

Critical point is a point no distinction in there between ~~Saturated liquid~~ and Saturated Vapor. Stage or the latent heat of vaporisation is zero.

Temperature Scale



Type of scale	Lower fixed point	Upper fixed point	Fundamental interval
Celsius ($^\circ C$)	$0^\circ C$	$100^\circ C$	$100^\circ C$
Fahrenheit ($^\circ F$)	$32^\circ F$	$212^\circ F$	$180^\circ F$
Absolute or Kelvin (K)	$273 K$	$373 K$	$100 K$
Raman (R)	$0 R$	$80 R$	$80 R$

To specify a scale of temperature the practice is to choose the temperature of the coldest substance available. In the past this was achieved by using the melting point of pure ice at a pressure of one standard atmosphere and marking

off the length of the mercury column in a glass capillary tube in thermal contact with the mixture as the lower point. The upper point is fixed by the length of a mercury column in a glass capillary tube placed in thermal contact with steam.

To determine any temperature T in degree Celsius at which the physical length of the mercury column in the thermometer is, X ,

$$T = \frac{X - X_0}{X_{100} - X_0} \times 100 \quad \text{--- (1)}$$

Where X_{100} = Steam point and X_0 = ice point.

To determine a kelvin temperature T at which the physical property has the value X ,

$$T = 273.16 \frac{X}{X_{tr}} \quad \text{--- (2)}$$

Where X_{tr} = the physical property at the triple point of water.

Constant Volume gas thermometers containing hydrogen or helium have excellent advantages in that they both have large coefficients of expansion which make small temperature changes easier to read as well as very low liquefaction points which provide a wide scale range. The Celsius Scale is defined as a temperature of 0.01°C to the triple point of water.

$$T(\text{Celsius}) = T(\text{kelvin}) - 273 \quad \text{--- (3)}$$

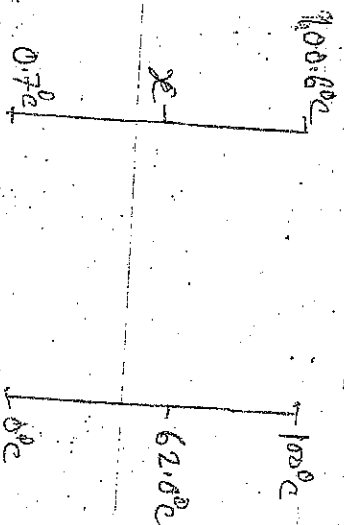
On this scale, the temperature at the ice point is found to be 0.01°C and at steam point to be 100.00°C . It is obviously an advantage for a thermometer to have a wide range to make it a versatile instrument. In any particular range, it is necessary to choose the best thermometer for sensitivity & accuracy, suitability and cost-effectiveness. Some different thermometers are described in the below Table (3).

Table (3)		
Types	Range	Use
Gas	$0 - 90^{\circ}\text{K}$	Low temperature, research.
Liquid-in-glass	$150 - 700^{\circ}\text{K}$	Wide range of general and specialised uses, eg clinical, meteorological.
Platinum resistance	$90 - 900^{\circ}\text{K}$	Standardising thermometers fixed point. Accurate determination in scientific works.
Thermocouple	$70 - 2000^{\circ}\text{K}$	Wide range of industrial uses. Standardising work between 900°K and 1300°K .
Thermistors	$200 - 1000^{\circ}\text{K}$	Numerous industrial measurement and control devices.

Examples

- (1) A thermometer is found to indicate 0.7°C at the lower fixed point and 100.6°C at the upper fixed point. What does this thermometer read when the true temperature is 62.0°C ?

Solution



$$x = ?, y = 62.0^\circ\text{C}, X_{100} = 100.6^\circ\text{C} \text{ and } X_0 = 0.7^\circ\text{C}$$

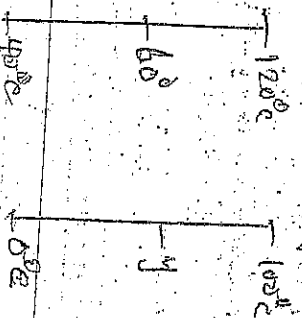
$$y = \frac{X - X_0}{X_{100} - X_0} \cdot X_{100} \text{ and } X = \frac{y}{100} (X_{100} - X_0) + X_0$$

$$X = \frac{62}{100} (100.6 - 0.7) - 0.7$$

$$X = 0.66 (101.3) - 0.7$$

$$\underline{\underline{X = 62.1^\circ\text{C}}}$$

- (2) The temperature of the melting point of ice and that of steam above water boiling at 760 mm mercury pressure are marked at 40°C and 120°C respectively on a certain thermometer. What is (i) the temperature in $^\circ\text{C}$ when the reading on this thermometer is 60° ? (ii) the thermometer reading when the temperature is 60°C ?



Solution

$$x = 60^{\circ}\text{C}, y = ?, x_{100} = 120^{\circ}\text{C}, x_0 = 40^{\circ}\text{C}$$

$$(i) \therefore y = \frac{x - x_0}{x_{100} - x_0} \times 100$$

$$y = \frac{60 - 40}{120 - 40} \times 100$$

$$y = \frac{20}{80} \times 100 = \underline{\underline{25^{\circ}\text{C}}}$$

$$(ii) x = \frac{y}{100} (x_{100} - x_0) + x_0$$

$$= \frac{60}{100} (120 - 40) + 40$$

$$= 0.6(80) + 40$$

$$= \underline{\underline{88^{\circ}\text{C}}}$$

- 3) The resistance of a platinum resistance thermometer is 10.4 ohms at 100°C . What is the resistance of the platinum thermometer when the temperature is 10.30 ohms ?

Solution

$$\therefore \theta = \text{Resistance at increased temperature} - \text{Resistance at } 0^{\circ}\text{C} \times 100$$

$$\text{Resistance at } 100^{\circ}\text{C} - \text{Resistance at } 0^{\circ}\text{C}$$

$$\theta = \frac{10.3 - 10.0}{10.4 - 10.0} \times 100$$

$$= \frac{0.3}{0.4} \times 100 = \underline{\underline{75^{\circ}\text{C}}}$$

C.G.S — Centimeter Gram Second System
F.P.S — Foot pound Second System
M.K.S — Meter kilogram second system
S.I. — System International

Temperature Conversion

(i) To Convert between Fahrenheit (°F) and degrees Celsius (°C).

$$T_C = \frac{5}{9} \times (T_F - 32) \quad \text{--- (i)}$$

$$T_F = \frac{9}{5} \times (T_C + 32) \quad \text{--- (ii)}$$

Where T_C is temperature in Celsius
" " " Fahrenheit.

(ii) To Convert between degrees Fahrenheit (°F) and Kelvin (K).

$$T_F = \frac{9}{5} \times (T_K - 273.15) + 32 \quad \text{--- (i)}$$

$$T_K = \frac{5}{9} \times (T_F - 32) + 273.15 \quad \text{--- (ii)}$$

Where T_F = Temperature in Fahrenheit
" " " Kelvin

(iii) To Convert between degrees Fahrenheit (°F) and Rankine (R)

$$T_R = T_R - 459.69 \quad \text{--- (i)}$$

$$T_C = T_R + 459.69 \quad \text{--- (ii)}$$

Where T_F is temperature in Fahrenheit
" " " Rankine

(iv) To Convert between degree Celsius (°C) and Kelvin (K)

$$T_C = T_K - 273.15 \quad \text{--- (i)}$$

$$T_K = T_C + 273.15 \quad \text{--- (ii)}$$

Where T_C is temperature in Celsius
" " " temperature in Kelvin

(v) To Convert between degrees (°C) and Rankine (R)

$$T_C = \frac{5}{9} \times (T_R - 459.69) - 32 \quad \text{--- (i)}$$

$$T_R = \left(\frac{9}{5} \times T_C + 32 \right) + 459.69 \quad \text{--- (ii)}$$

where T_c is temperature in Celsius,

T_R = Temperature in Rankine.

(vi) To convert between degree Kelvin (K) and Rankine (R).

$$T_K = \left(\frac{5}{9} \times (T_R - 459.69) - 32 \right) + 273.15 \quad \text{--- (viX)}$$

$$T_R = \left[\frac{9}{5} \times (T_K - 273.15) + 32 \right] + 459.69 \quad \text{--- (viX)}$$

Where T_K is temperature in Kelvin
 T_R is " " " " Rankine

Examples

① Convert 37°C to both Fahrenheit degree and Kelvin.

$$T_F = \left(\frac{9}{5} \times T_C + 32 \right) = \frac{9(37) + 32}{5} = 98.6^\circ\text{F} \quad (\text{that is human body temperature}).$$

$$T_K = T_C + 273.15 = 37 + 273.15 = \underline{310.15\text{K}}$$

② Convert 80K to both Celsius and Fahrenheit degrees.

Solution

$$T_C = T_K - 273.15 = 80 - 273.15 = \underline{-193.15^\circ\text{C}}$$

$$T_F = \frac{9}{5} T_C + 32 = \frac{9(-193.15) + 32}{5} = -347.7^\circ\text{F}$$

③ Calculate the difference between 97°F and 40°F in both Celsius degree and Kelvin in Celsius degrees. These temperature are!

Solution

$$T_C = \frac{5}{9} (T_F - 32) = \frac{5}{9} (97 - 32) = 37.22^\circ\text{C}$$

$$T_C = \frac{5}{9} (40 - 32) = 4.44^\circ\text{C}$$

The difference in Celsius is $37.22 - 4.44 = 32.78^\circ\text{C}$

$$T_K = T_C + 273.15 = 32.78 + 273.15 = \underline{306\text{K}}$$

The difference in Kelvin is 306K.

④ The boiling temperature of liquid oxygen (O_2) is -297.3°F . Convert this temperature to Celsius degrees and Kelvin.

$$T_C = \frac{5}{9} (-297.3 - 32) = -182.9^\circ\text{C} \quad T_K = -182.9 + 273.15 = \underline{90.2\text{K}}$$

Calorimetry

When two bodies are in contact with each other, the thermal equilibrium will establish due to transfer of heat from higher temperature T_H to a lower temperature T_L .

$$T_L < T_E < T_H \quad \text{--- (i)}$$

where T_E is equilibrium temperature.

The fundamental principle of Calorimetry states that, heat lost by a body at higher temperature equal to heat gained by the body at lower temperature.

$$Q_{\text{lost}} = Q_{\text{gained}} \quad \text{--- (ii)}$$

The unit of heat in C.G.S system is Calorie and in S.I unit is Joules.

Factors affecting the quantity of heat

(1) The quantity of heat (Q) absorbed by a body is directly proportional to the mass (m) of the body. $Q \propto m$ --- (iii)

(2) The quantity of heat (Q) absorbed by a body is directly proportional to rise in temperature (ΔT).

$$Q \propto \Delta T \quad \text{--- (iv)}$$

By adding the two equations (iii & iv) together we have, $Q \propto m \Delta T$

$$\text{and } Q = m S \Delta T \quad \text{--- (v)}$$

where S is specific heat of the body (Joules)

The unit of heat in CGS system is Calorie (Cal)

$$1 \text{ Calorie} = 4.18 \text{ joules}$$

$$1 \text{ K Calorie} = 4.18 \times 10^3 \text{ joules}$$

The quantity of heat (Q) absorbed or released by a body comes from material of that material because of the characteristics of that material. This is called "Specific heat" of the material.

$$Q = ms \Delta T \text{ cal} \quad \text{--- (1)}$$

$$S = Q / m \Delta T$$

ΔT = The rise or fall in temperature.

S = Specific heat of the material.

When $m = 1$ and $T = 1^\circ\text{C}$

$$\text{We get } S = Q \quad \text{--- (2)}$$

Thus, CGS system unit of $S = \text{Cal/g}^\circ\text{C}$ and SI unit is $\text{J/kg}^\circ\text{C}$.

Thermal heat capacity is defined as the quantity of heat energy required or needed to raise the temperature of a substance by 1°C . It is measured in J/K .

Specific heat capacity of a substance is defined as the quantity of heat required to raise the temperature of a unit mass of a substance by 1°C or 1°K . It is denoted by C_s , measured in $\text{J/kg}^\circ\text{K}$. It is a scalar quantity.

Example

- ① When a piece of iron of mass 4 kg and specific heat capacity $460 \text{ J/kg}^\circ\text{C}$ cools from 80°C to 20°C . Calculate heat given out.

Solution

$$m = 4 \text{ kg}, c = 460 \text{ J/kg}^\circ\text{C}, \Delta\theta = 80^\circ\text{C} - 20^\circ\text{C} = 60^\circ\text{C}$$

$$\begin{aligned} Q &= mc\Delta\theta \\ &= 4 \times 460 \times 60 = 110,400 \text{ J} \\ &\approx 110.4 \text{ kJ} \end{aligned}$$

Methods of determining the specific heat capacity of a substance

- ① Solid by the method of mixtures.

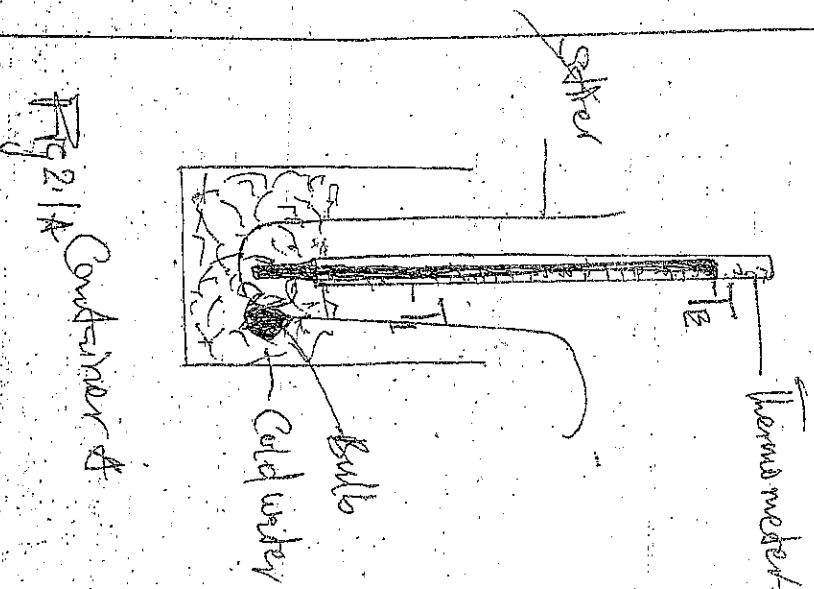


Fig 2.1A Container &

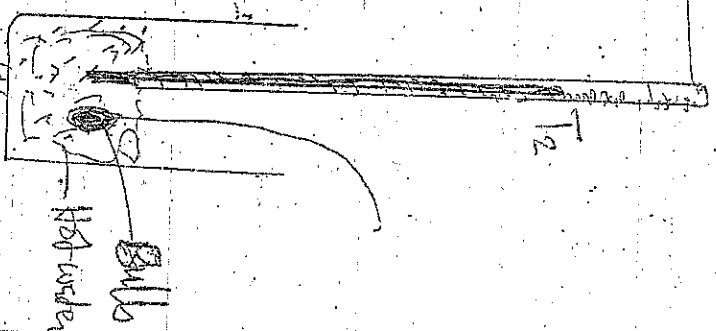


Fig 2.1B Heat source
Container B.

Heat the solid body (bulb) in hot water at temperature T_2 from Container (B). Quickly transfer it into a second Container (A) (well-lagged) that containing cold water at temperature T_1 . Stir it gently and record the final temperature (equilibrium temperature) T_E .

Assuming the energy lost to the surrounding and the Container to be negligible. We express as, Heat lost by the solid = Heat gain by the liquid.

$\Phi_{\text{lost by the solid}} = \Phi_{\text{gained by water}}$

$$MS_S(T_2 - T_E) = MS_W(T_E - T_1) \quad \text{--- (i)} \quad (V \text{ vol})$$

$$S_S = \frac{MS_W(T_E - T_1)}{M \times (T_2 - T_E)} \quad \text{--- (ix)}$$

where S_S = Specific heat of solid (bulb)
 S_W = " " " " liquid (water).

(2) Liquid by the method of mixtures

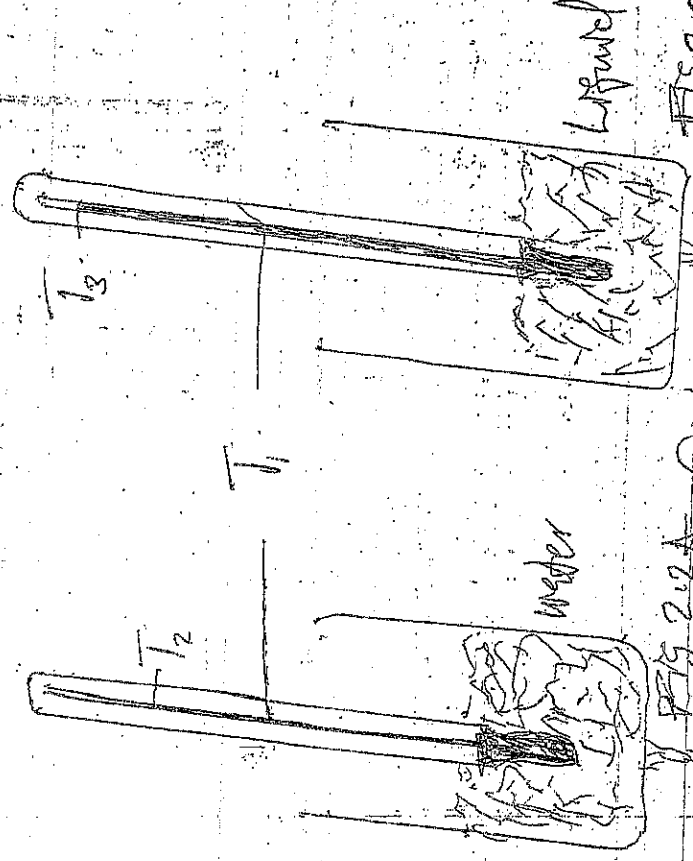


Fig 2.2A (B)

Fig 2.2B

filled two different containers with equal masses with different substances (water and liquid). The initial temperature of both substances is T_1 and after heating both the containers separately at least 10 minutes. Record the temperature of water as T_2 and that of liquid as T_3 .

Heat gained by water = Heat gained by liquid.

$$(m_s \Delta T)_{\text{water}} = m_s \Delta T_{\text{liquid}}$$

$$m_s m_w (T_2 - T_1) = m_s (T_3 - T_1)$$

∴ mass of water = mass of liquid

$$S_w (T_2 - T_1) = S_l (T_3 - T_1)$$

$$S_l = S_w \frac{(T_2 - T_1)}{(T_3 - T_1)} \quad \text{--- (X)}$$

$$\text{or } S_l = \frac{(T_2 - T_1)}{(T_3 - T_1)} \quad \therefore S_w = 1 \quad \text{--- (XII)}$$

③ Solid by the electric method.

An electric heater is inserted in a hole drilled in the solid block, and current is passed through it to give heat to the block. The solid (metal block) is insulated from the surroundings. The temperature change in the solid is found by means of a thermometer placed in another hole bored in the solid.

Heat given by heater = VIt ,
where V is the voltage supply which passes
current of I amperes for time (second).

Heat received by the solid = $mS\Delta\theta$

where m is the mass of the solid material
 block, S is specific heat capacity and
 $\Delta \theta$ is change in temperature.

$$\therefore m S \Delta \theta = V I t$$

$$S = \frac{V I t}{m \Delta \theta} \quad \text{--- (XII)}$$

Specific heat capacity of some substances

in cal/gm°C.

Normal water	1 cal/gm°C
Sea water	0.94 cal/gm°C
Mercury	0.03 cal/gm°C
Paraffin oil	0.52 cal/gm°C
Olive oil	0.47 cal/gm°C
Turpentine	0.42 cal/gm°C
Thyle alcohol	0.6 cal/gm°C
Glycerin	0.58 cal/gm°C
Aluminium	0.22 cal/gm°C
Copper	0.09 cal/gm°C
Ice	0.5 cal/gm°C
Iron	0.12 cal/gm°C
Lead	0.03 cal/gm°C

1 cal
= 4.18 J

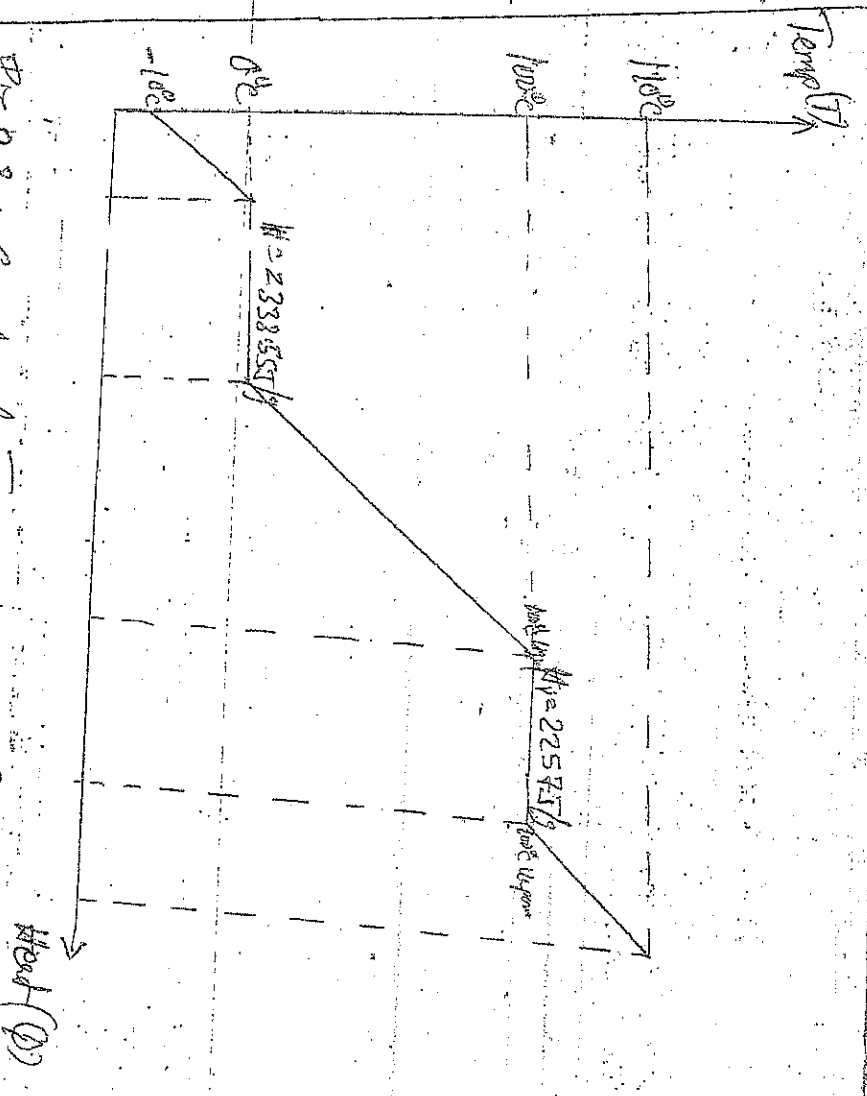


Fig 2.13 graph of Temperature against heat (Q).

$H_f = 333.55 \text{ J/g}$ $H_v = 2257 \text{ J/g}$

Specific heat of water (H_2O) solid $\approx 2.05 \text{ J/gK}$

liquid $\approx 4.178 \text{ J/gK}$

vapor $\approx 1.871 \text{ J/gK}$

The amount of heat require to rise & deapre
- volume from -10°C ice $\rightarrow 110^\circ\text{C}$ vapor of 200g.

-10°C ice $\rightarrow 0^\circ\text{C}$ ice $\approx 2.05 \text{ J/g}^\circ\text{C} \times 200 \text{g} \times 10^\circ\text{C}$
 $\approx 4100 \text{ J}$

0°C ice $\rightarrow 0^\circ\text{C}$ water $\approx 333.55 \text{ J/g} \times 200 \text{g} \approx 66710 \text{ J}$

0°C water $\rightarrow 100^\circ\text{C}$ water $\approx 4.178 \text{ J/g}^\circ\text{C} \times 200 \text{g} \times 100^\circ\text{C}$
 $\approx 83560 \text{ J}$

100°C water $\rightarrow 100^\circ\text{C}$ vapor $\approx 2257 \text{ J/g} \times 200 \text{g} \approx 451400 \text{ J}$

100°C vapor $\rightarrow 110^\circ\text{C}$ vapor $\approx 1.87 \text{ J/g}^\circ\text{C} \times 200 \text{g} \times 10$
 $\approx 3780 \text{ J}$

Total heat $\approx 4100 \text{ J} + 66710 \text{ J} + 83560 \text{ J} + 451400 \text{ J} + 3780 \text{ J}$
 $\approx 609950 \text{ J}$

(21) Check the above figures.

② The temperature of 400g of a metal is reduced from 21°C to 100°C and it is then placed in 200g of water at 10°C . Calculate the specific heat capacity of metal. [Specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{K}^{-1}$].

Solution

$$M_1 = 400\text{g} \approx 0.4\text{kg}, \theta_1 = 21^{\circ}\text{C}, \theta_2 = 100^{\circ}\text{C}, m_2 = 200\text{g} \approx 0.2\text{kg}$$

$$\theta_3 = 10^{\circ}\text{C} \text{ and } C_w = 4200 \text{ J/kgK, and } C_m = ?$$

$$\text{Heat given out by metal} \approx 0.4 \times C_m \times (100 - 21)$$

$$\text{Heat received by water} \approx 0.2 \times 4200 \times (21 - 10)$$

$$\therefore C = \frac{0.2 \times 4200 \times 11}{0.4 \times 79} = \underline{\underline{292.4 \text{ J/kgK}}}$$

③ 30g of a metal at 90°C is placed in 70g of water at 20°C in a calorimeter of mass 70g made of the same metal. If the final temperature is 22.1°C . Calculate the specific heat capacity of the metal.

Solution

$$Q = mC\theta$$

$$\text{Heat lost by hot metal} \approx 0.03 \times C \times (90 - 22.1)$$

$$\text{Heat gained by cold water} \approx 0.070 \times 4200 (22.1 - 20)$$

$$\text{Heat energy lost} = \text{Heat energy gained}$$

$$0.03 \times C \times (90 - 22.1) \approx 0.070 \times 4200 (22.1 - 20)$$

$$C = \underline{\underline{420 \text{ J/kgK}}}$$

④ A 100W heater causes the temperature of 4kg block of metal to rise from 15°C to 57°C in 45 min. What is the specific heat capacity of the metal?

Solution

$$\text{Energy supplied by heater} = \text{Heat energy supplied per sec}$$

$$100 \times (45 \times 60) \approx 4 \times C \times (57 - 15)$$

$$C \approx \underline{\underline{500 \text{ J/kgK}}}$$

(5) How long does it takes a 750W heater operating at full rating to raise the temperature of 1kg of water from 40°C to 70°C . (Specific heat capacity of water is equal to 4200 J/kg and neglect heat losses).

Solution

$$P = 750\text{W}, \Delta\theta = \theta_2 - \theta_1 = 30^\circ, m = 1\text{kg}$$

$$Pt = mc\Delta\theta$$

$$t = \frac{4200 \times 30}{750} = \underline{\underline{168\text{ sec}}}$$

(6) Hot water at temperature of 75°C is added to twice that amount of water at a temperature of 30°C . If the resulting temperature of the mixture is 50°C , Calculate t .

Solution

Let the mass of hot water be m and mass of cold water be $2m$.

Heat lost by hot water = Heat gained by Cold water

$$(mc\Delta\theta)_{\text{hot water}} = (mc\Delta\theta)_{\text{cold water}}$$

$$\therefore mc(t - 50) = 2m \times c \times (50 - 30)$$

$$t - 50 = 20$$

$$t = 40 + 50 = \underline{\underline{90^\circ\text{C}}}$$

(7) A piece of Copper metal, 200g is removed from a furnace at a temperature of 80°C and quickly dropped into a Copper vessel of mass 100g containing 100g of water. If the initial temperature is 30°C . Find the value of θ . (Assume S.H.C of Copper = $4200\text{ J/kg}^\circ\text{C}$ and water = $4200\text{ J/kg}^\circ\text{C}$).

Solution

Heat lost by Copper metal = Heat gained by vessel and water

(23)

$$m_c C (\theta - 70) = m_i C (70 - 20) + m_w C (70 - 30) \\ 200 \times 4200 (\theta - 70) = (100 \times 4200 \times 40) + (100 \times 4200 \times 40) \\ 200 \times 4200 (\theta - 70) = 168000 + 1680000$$

$$\theta - 70 = \frac{1696800}{200 \times 4200}$$

$$\theta - 70 = 202$$

$$\theta = 202 + 70$$

$$\theta = \underline{\underline{272^\circ C}}$$

Check please

LATENT HEAT

Latent heat is defined as the quantity of heat required or needed to change the state of a substance at constant temperature and pressure. It is a scalar quantity.

$$\text{Thus } Q = mL \text{ --- (XIV)}$$

m = mass

L = specific latent heat

Specific latent heat of fusion (ice)

We consider the change of state from solid to liquid called Fusion.

Specific latent heat of fusion (ice) is defined as the quantity of heat required or needed to change a unit mass of solid (ice) into a liquid (water) at constant temperature. Melting point measured in $^\circ C$ or $^\circ F$. It is also a scalar quantity.

The experimental value of $L_{ice} = 336 \text{ J/g}$

Thus, the experimental value of $L_{ice} = 336 \text{ J/g}$

Q. 0.5 kg of water at $10^\circ C$ is completely converted to ice at $0^\circ C$ by extracting 18800 J of heat from it. If the specific heat capacity of water is $4200 \text{ J/kg} \cdot ^\circ C$, calculate the specific latent heat of fusion of ice.

Solution

$$Q = mL_{ice} + MC_{AB}$$

$$188000 = 0.5L + 0.5 \times 4200 \times (10 - 0)$$

$$188000 = 0.5L + 21000$$

$$0.5L = 188000 - 21000$$

$$0.5L = 167000$$

$$L = \frac{167000}{0.5} = \underline{\underline{334000 \text{ J/kg}}}$$
 Check

(c) What amount of current would pass through a 1000 coil if it takes 21 sec for the coil to just melt a lump of ice of mass 10g at 0°C . If there are no heat losses? ($L_{ice} = 336 \text{ J/g}$).

Solution

$$P \times t = ML, \quad P = I^2 R$$

$$\therefore I^2 R \times t = ML$$

$$I^2 \times 10 \times 21 = 10 \times 336$$

$$210 I^2 = 3360$$

$$I^2 = \frac{3360}{210} = 16$$

$$I = \sqrt{16} = \underline{\underline{4 \text{ A}}}$$

Check.

Melting point of a substance is defined as the temperature at which the substance changes from the solid to the liquid state.

Specific Latent Heat of Vaporisation (Steam)

The change of state from liquid to vapour is called vaporisation.

Specific latent heat of vaporisation (steam) is defined as the quantity of heat required or needed to change a 'unit mass' of liquid to gaseous state at constant temperature (boiling point) and pressure. It is denoted by h_{steam} .

measured in J/kg or J/g and a scalar quantity.
Thus, the experimental value of $L_{\text{steam}} = 2260-226$

J/kg. Hence, $L_{\text{ice}} = \frac{1}{7} L_{\text{steam}}$. ——— (xiv)

(9) 0.5 kg of a certain substance contained a copper can of mass 0.5 kg is melted in a water bath at 80°C and raised to a temperature of 100°C . Calculate the total heat given out when the can and its contents are allowed to cool to room temperature, 20°C (melting point of the substance is 80°C and the specific heat capacity for both liquid and solid is 2100 J/kgK , specific latent heat of fusion is 170000 J/kg , specific heat capacity of copper is 400 J/kgK).

Solution

Neglecting all possible losses, the heat given out by the copper can in cooling from 100°C to 20°C is $0.4 \times 400 \times 80 = 12800 \text{ J}$

Heat given out by the substance of fusion at 80°C is $0.5 \times 170000 = 85000 \text{ J}$

Heat given out by the solid substance on fusion at 80°C to 20°C is $0.5 \times 2100 \times 60 = 63000 \text{ J}$
Total heat given out $= 12800 + 21000 + 85000 = 128180 \text{ J}$

(10) A 100 g mass of metal is just melted by heating it to 500°C . It is then poured to a tray of water at 20°C . If the final temperature of the water is 40°C and the specific heat capacity of metal is 200 J/kgK , calculate its specific latent heat of fusion.

(20)

Solidifying

Heat lost = Heat lost in solidifying + heat lost in cooling.
= Heat energy gained by water

$$0.20 \times L \times 0.20 \times 4400 \times (500 - 460) = 2 \times 442000 \times 20$$

$$L = \frac{2 \times 442000 \times 20 \times 0.20 \times 4400 \times (460)}{0.20}$$

$$L = 656 \text{ KJ/kg}$$

Vapour (Change of state)

Boiling is a process that occurs throughout the entire volume of a liquid before reaching a certain boiling point. Thus, boiling point is defined as the temperature of the saturated vapour pressure (S.V.P) of the liquid is equal to its external (atmospheric) pressure.

Therefore this is saturated and an unsaturated vapour.

Saturated vapour.

Unsaturated vapour

(1) Saturated vapour is in contact with its own liquid. Unsaturated vapour is not in contact with its own liquid.

(2) The pressure of a saturated vapour is independent of its volume. The pressure of an unsaturated vapour is approximately inversely proportional to its volume.

Saturated

Boils are directly proportional to its absolute temperature.

Note (1) Unsaturated vapour in an unsaturated vapour is one which is not in contact with its own liquid.

(2) Saturated vapour is one which is in a state of dynamic equilibrium.

The factors affecting boiling are as follows.

- (i) Impurities (ii) nature of liquid i.e., density
- (iii) Inter molecular force (iv) pressure.

Evaporation

Evaporation is a process which involves the escape of molecules from the surface of liquid to vapour (gaseous state). Cooling is always accompanied with evaporation because the average K.E. of the molecules of liquid remaining is reduced.

The factors affecting evaporation are as follows.

- (i) Temperature (ii) nature of the exposed surface area of liquid (iii) nature of liquid (density).
- (iv) Wind and dryness.

Difference b/w Boiling and Evaporation

Boiling

(i) Boiling takes place at a definite temperature. It takes place at all temperatures.

(ii) During boiling, temperature is constant.

During evaporation, temperature varies.

(iii) Vapour escapes both from the surface and interior of the liquid. Vapour escapes only from the liquid surface.

(iv) Boiling requires a considerable amount of heat. Evaporation requires less heat.

Similarities

(i) In both, vapour escapes from the surface.

(ii) Both are affected by pressure.

(iii) Both depend on the nature of the liquid (density).

Volatile Liquids

Volatile liquids such as methylated spirit and other evaporate very easily due to their low boiling point. Thus, the latent heat needed to change the liquid to vapour is small, which is absorbed from the liquid at the same temperature.

Newton's Law of Cooling

Newton's Law of Cooling states that the rate at which a hot body loses heat is directly proportional to the difference between the temperature of the hot body and that of its ambient temperature (temperature of surrounding).

$$\frac{dT}{dt} \propto (T_H - T_A)$$

$$\frac{dT}{dt} = -k(T_H - T_A) \quad \text{--- (1)}$$

Where k is the constant of proportionality. To solve the full eqn, we have to multiply both of the eqn (1) with dt , we have,

$$dT = -k(T_H - T_A) dt$$

$$\frac{dT}{T_H - T_A} = -k(T_H - T_A) dt$$

$$\frac{dT}{(T_H - T_A)} = -k \frac{(T_H - T_A)}{(T_H - T_A)} dt$$

$$dT = -k dt \quad \text{--- (2)}$$

$$(T_H - T_A)$$

Applying integration to both side of eqn (2)

$$\int \frac{1}{(T_H - T_A)} dT = \int -k dt \quad \text{Recall } \int \frac{1}{x} = \ln(x) \text{ and } \int dx = x$$

$$\ln(T_H - T_A) = -kt + C$$

$$e^{\ln(T_H - T_A)} = e^{-kt + C}$$

(29)

$$e^{\ln \left(\frac{T_H - T_A}{T_H - T_A} \right)} = e^{-kt} e^{-kt}$$

$$T_H - T_A = C e^{-kt}$$

$$T(t) = T_A + C e^{-kt} \quad (ii)$$

Let use eqn (ii) and find C , at $t=0$.

$$T(0) = T_A + C e^{-k(0)}$$

$$T(0) = T_A + C (e^0)$$

$$T_H = T_A + C$$

$$C = T_H - T_A \quad (iii)$$

Substitute eqn (iii) into (ii), we have,

$$T(t) = T_A + (T_H - T_A) e^{-kt} \quad (v)$$

Where $T(t)$ = Temperature of time t .

T_A = Ambient temperature (temperature of -

Surroundings).

T_H = Temperature of hot object at time (0)

k = positive constant

t = time

Ex sample

If takes 12 min for an object at 100°C to cool to 80°C in a room at 50°C . How much longer will it take for its temperature to decrease to 70°C ?

Soln

$$T(t) = T_A + (T_H - T_A) e^{-kt}$$

$$T(12) = 50 + (100 - 50) e^{-12k}$$

$$80 = 50 + (100 - 50) e^{-12k}$$

$$80 - 50 = 50 e^{-12k}$$

$$30 = 50 e^{-12k}$$

$$\begin{array}{l} \text{1st data} \\ t = 12 \text{ min} \quad T_{12} = 80^\circ\text{C} \end{array}$$

$$T_A = 50^\circ\text{C}$$

$$T_H = 100^\circ\text{C}$$

$$k = ?$$

$$\text{2nd}$$

$$T(t) = 70^\circ\text{C}$$

$$T_H = 50^\circ\text{C}$$

$$T_H = 80^\circ\text{C}$$

$$30 = 50 e^{-12k}$$

divide both side by 50, we have

$$\frac{30}{50} = \frac{50}{50} e^{-12k}$$

$$\frac{3}{5} = e^{-12k}$$

$$\ln\left(\frac{3}{5}\right) = \ln(e^{-12k})$$

$$\ln\left(\frac{3}{5}\right) = -12k \ln e$$

$$\ln\left(\frac{3}{5}\right) = \frac{-12k}{\cancel{\ln e}}$$

$$k = 0.04257$$

Substitute k in eqn (v), we have

$$T(t) = T_A + (T_H - T_A) e^{-kt}$$

$$70 = 50 + (80 - 50) e^{-0.04257t}$$

$$70 - 50 = 30 e^{-0.04257t}$$

$$20 = 30 e^{-0.04257t}$$

divide both side by 30, we have

$$\frac{20}{30} = \frac{30}{30} e^{-0.04257t}$$

$$\frac{2}{3} = e^{-0.04257t}$$

$$\ln\left(\frac{2}{3}\right) = \ln(e^{-0.04257t})$$

$$t = \frac{\ln\left(\frac{2}{3}\right)}{-0.04257} \approx 9.52 \text{ min}$$

$$\begin{array}{c} 100^{\circ}\text{C} \xrightarrow[12\text{ min}]{80^{\circ}\text{C}} 70^{\circ}\text{C} \\ \boxed{T_A = 50^{\circ}\text{C}} \quad \quad \quad 21.52\text{ min} \quad \quad \quad 9.52\text{ min} \end{array}$$

To prove it

$$T(t) = T_A + (T_{\infty} - T_A)e^{-kt}$$

$$70 = 50 + (100 - 50)e^{-0.04257t}$$

$$70 - 50 = 50e^{-0.04257t}$$

$$20 = 50e^{-0.04257t}$$

$$20 = 50e^{-0.04257t}$$

$$\frac{20}{50} = \frac{50}{50}e^{-0.04257t}$$

$$\ln\left(\frac{2}{5}\right) = \ln e^{-0.04257t}$$

$$\ln\left(\frac{2}{5}\right) = -0.4257t$$

$$t = \frac{\ln\left(\frac{2}{5}\right)}{-0.04257} = \underline{\underline{21.52\text{ min}}}$$

Kinetic Theory of Gas

Kinetic theory considers matter to be composed of discrete particles (atoms and/or molecules) in continuous motion. In a gas, the molecules are in random motion with a wide distribution of speed ranging from zero to very large values.

Atoms: A unit of matter, the smallest unit of a chemical element. Each atom consists of nucleus which has a positive charge, and a set of electrons that move around it.

Molecules: A group of two or more atoms that form the smallest identifiable unit into which a pure substance can be divided and still retain the composition e.g. H_2O .

Avogadro's Number/Constant: (N_A) is the number of particles (molecules or atoms) in 1 kmol of substance.

For all substances, $N_A = 6.022 \times 10^{26}$ particles/kmol

$\approx 6.022 \times 10^{23}$ particles/mol (molecules) for bond is about $\times 10^{23}$ molecules, $M = 2 \text{ kg/kmol}$ of H_2 gas in standard

therefore $2 \text{ kg of } H_2$ and $32 \text{ kg of } O_2$ each containing 6.022×10^{26} molecules.

Relative molecular mass/mass of a molecule (or atom)

Can be found from the molecular (or atomic) mass (M) of the substance and Avogadro's number (N_A). Since M kg of substance contains N_A particles, the mass of one particle is given by $\frac{M}{N_A}$.

$$m = \frac{M}{N_A}$$

Mole: is the unit of amount of substance with international system of units. The quantity which the substance is a measure of containing etc.

object or sample.

Molar mass: Is defined as the mass in grams of one mole of a substance. The units of molar mass are grams per mole abbreviated as g/mol . The mass of a single isotope of any given element (the isotopic atomic mass) is a value relating the mass of that isotope to the mass of the isotope Carbon 12.

Molar Volume at S.T.P. Is the Volume occupied by one mole of a chemical element or a chemical compound. It can be calculated by dividing the molar mass (M) by the mass density (ρ).

S.T.P. Standard Condition is Standard Temperature and pressure (S.T.P.) are defined to be

$$T = 273.15 \text{ K} = 0^\circ \text{C}, P = 1.013 \times 10^5 \text{ Pa}$$

Under standard condition, 1 kmol of ideal gas occupies a volume of 22.4 m^3 . Therefore, at S.T.P., 2 kg of H_2 occupies a

the same volume as 32 kg of O_2 or 28 kg of N_2 , namely 22.4 m^3 .

Differentiate between number of molecules of the gas and molar volume, Number of molecules and Avogadro's constant.

Number of mole is a measure of the large unit for the amount of substance present in a given sample. It is a number that is equal to 6.022×10^{23} particles.

To determine the quantity of Sulphur (S_8) molecules present in a solid Sulphur of present in 16 g of solid Sulphur.

Avogadro's Constant is the number of atoms/molecules in a sample can be calculated by multiplying the number of moles by the Avogadro's Constant.

Number of mole $\times N_A$
1 mole of solid Sulphur S_8 is $8 \times 32 \text{ g} = 256 \text{ g}$

256g of solid Sulphur contains $= N_A$ molecules.

$$\text{kg of Solid Sulphur contains} = \frac{6.022 \times 10^{23}}{256} \times 16 \text{ molecule}$$

$$= 3.76375 \times 10^{22} \text{ molecules.}$$

Avogadro's number was originally defined as the number of molecules in one gram of hydrogen. The number was then redefined to be the number of atoms in 12 grams of Carbon 12 isotope, Carbon-12. The results of many experiments have determined that there are $6.02214129 \times 10^{23} \pm 0.00000027 \times 10^{23} \pm 6.02214129 \times 10^{23}$ molecules in one mole of Carbon 12 atoms.

Recall that the mole is 18 base unit in the SI system of units that is a unit for the amount of substance. [mole]. The mole is defined as the amount of substance any substance that contains as many atoms as there are in 12 grams of Carbon 12. The number of molecules per mole is called the Avogadro Constant and is $N_A = 6.0221415 \times 10^{23} \text{ mol}^{-1}$. $52 = 31173$

Determine the number of mole in 95g of MnO_2 .

Soln

$$\text{Mass of } \text{MnO}_2 = 95\text{g}$$

$$\text{mass of one mole } \text{MnO}_2 = 86.94\text{g}$$

$$\text{Number of mole} = \frac{\text{Mass of substance}}{\text{mass of one mole}} = \frac{95}{86.94} = 1.092\text{mol}$$

$$\text{Mass of one mole} = 86.94$$

Molar Volume: Avogadro's law tells us that equal volume of any gas at the same temperature and pressure contains an equal number of particles. The independent variable in the above case is the number of particles. That one mole of Chlorine gas, Cl_2 , contains the same number of particles as one mole of Hydrogen gas, H_2 , under identical conditions of pressure and temperature. Should occupy the same volume.

Same Volume the molar volume is the volume occupied by one mole of a gas. It is the same for all gases.

Examples

(1) Converting from Amount to Volume

* party balloon has 2.5 mol of helium gas.
What is the volume of the balloon at STP?

$$\frac{22.4 \text{ L}}{1 \text{ mol}} \times 2.5 \text{ mol} = 56 \text{ L}$$

$$V = n \times 22.4 \text{ L/mol} = 2.5 \text{ mol} \times 22.4 \text{ L/mol} = 56 \text{ L}$$

$$V = 2.5 \text{ mol} \times \frac{22.4 \text{ L}}{1 \text{ mol}} = 56 \text{ L}$$

(2) mass to volume
A sample of helium at STP has a mass of 32 g. What volume does this mass of gas occupy?

$$M_{\text{He}} = 32.0 \text{ g/mol} = 4.0 \text{ g/mol} \times 8 = 32.0 \text{ g/mol}$$

$$V = n \times \frac{22.4 \text{ L}}{1 \text{ mol}} = 8 \text{ mol} \times \frac{22.4 \text{ L}}{1 \text{ mol}} = 179.2 \text{ L}$$

Step 1: Determine the amount of helium gas in the sample by multiplying the mass of helium by its molar mass. Conversion factor derived from the molar mass of helium.

Step 2: Determine the volume of the gas in the sample by multiplying the amount of helium by its molar volume. Conversion factor derived from the molar volume of gas at STP.

$$\frac{1 \text{ mol}}{4.0 \text{ g}} \times 32.0 \text{ g} = 8.0 \text{ mol}$$

$$\frac{1 \text{ mol}}{22.4 \text{ L}} \times 179.2 \text{ L} = 8.0 \text{ mol}$$

مکتبہ اسلامیہ دارالعلوم دیوبند

سید علی محمد علی

Only

(1), my gas coming out of the perfectly elastic material in reaction of

(ii) Gas molecules do not react completely randomly.

thrust motion vs. compression /
fracture in strand line

(iv) The drive motor

perfectly elastic. It means

Considered
in such collision 15

(10) blue mofn

(vi) the effect of gravity on

Maxwell's law of distribution of velocity

✓ to find the number of
✓ (ie C to Cdc),
✓ law

When given this equation Maxwell did the following

(1) Speed of gas molecules ranges from zero to infinity

(b) In the spread state, the density of gas remains constant.

(20) Number of gas molecules have speed between the finite range.

Consider a g ses Cuspidal N number n in $\mathcal{N}$ and n is prime to g .
 If n is prime to g then n is prime to g and n is prime to g .

velocity c and $\hbar$ and ϵ composition. X , Y and Z are the three, true, probability of nucleon

thence from the probability theory, and in particular to the theory of the stochastic process.

having velocity component v_x to $v_{x\max}$
 having velocity component v_y to $v_{y\max}$
 having velocity component v_z to $v_{z\max}$

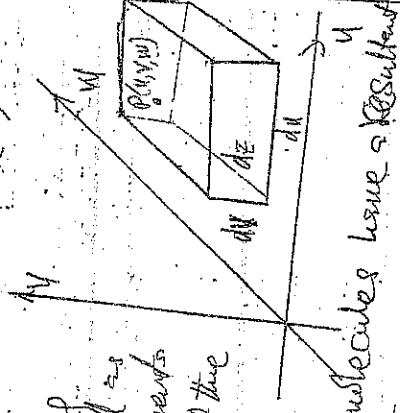
generality of procedures using networks
in $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ and the possibility of procedures

Component w to $w+dw$ $\rightarrow f(w)dw$. Thus the number of molecules whose velocity lies between u to $u+du$, v to $v+dv$ and w to $w+dw$ is given by:

$$dN = N f(u) du f(v) dv f(w) dw$$

$$dN = N f(u) f(v) f(w) du dv dw \quad (i)$$

If velocity components u , v , and w along the three axes, the speed formal is called as Velocity space. A molecule having components u, v, w be represented by a point P . Let the small volume du, dv, dw around P .



Contains dN number of molecules. This molecules have resultant velocity C is given by: $C^2 = u^2 + v^2 + w^2$. Differentiating on both side, we get

$$0 = 2u du + 2v dv + 2w dw \quad \therefore C = \text{Constant}$$

$$\therefore u du + v dv + w dw = 0 \quad \text{--- (ii)}$$

The probability density $f(u)f(v)f(w)$ should depend only on the magnitude but not the direction of C .

$\therefore f(u)f(v)f(w) = f(C^2)$ (where $f(C^2)$ is function of C^2). Differentiating on both side, we get

$$d[f(u)f(v)f(w)] = 0 \quad \therefore C^2 = \text{Constant}$$

$f'(w) du f(v) f(w) + f'(v) dv f(u) f(w) + f'(u) du f(v) f(w) = 0$. Dividing above eqn by $f(u)f(v)f(w)$, we get,

$$\frac{f'(u)}{f(u)} du + \frac{f'(v)}{f(v)} dv + \frac{f'(w)}{f(w)} dw = 0 \quad \text{--- (iii)}$$

Multiplying the eqn (iii) by an arbitrary constant (B) and adding to eqn (ii), we get:

$$\left[\frac{f'(u)}{f(u)} + Bu \right] du + \left[\frac{f'(v)}{f(v)} + Bv \right] dv + \left[\frac{f'(w)}{f(w)} + Bw \right] dw = 0$$

As du, dv and dw can not be zero independently, so we get, $\frac{f'(u)}{f(u)} + Bu = 0$, $\frac{f'(v)}{f(v)} + Bv = 0$, $\frac{f'(w)}{f(w)} + Bw = 0$

$$\frac{f'(u)}{f(u)} = -Bu$$

Integrating wrt u , we get $\log f(u) = -\frac{1}{2} B u^2 + \log a$. Taking anti-log on both side, we get (where $\log a$ is integrating constant).

$f(u) = a e^{-\frac{1}{2} B u^2} = a e^{-\frac{1}{2} u^2}$ (where, $B = -\frac{1}{2} B = \text{Constant}$)

$$(38)$$

Substituting $f(u)$, $f(v)$ and $f(w)$ in eq. (1) we get,

$$dN = N(a e^{-bu^2})(a e^{-bv^2})(a e^{-bw^2}) du dv dw$$

$$dN = N a^3 e^{-\frac{b}{2}(u^2+v^2+w^2)} du dv dw$$

$$C^2 = u^2 + v^2 + w^2$$

$$dN = N a^3 e^{-\frac{b}{2} C^2} du dv dw \quad \text{--- (iv)}$$

In this eqn constant a and b is found to be

$$a = \sqrt{\frac{m}{2\pi kT}} \quad \text{and} \quad b = \frac{m}{2kT}$$

Where m is the mass of the gas particles, k is Boltzmann Constant, T is temperature of the gas.

Then eqn (iv) becomes,

$$dN = N \left(\sqrt{\frac{m}{2\pi kT}} \right)^3 e^{-\frac{mC^2}{2kT}} du dv dw \quad \text{--- (v)}$$

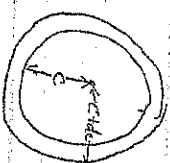
Eqn (v) is one form of Maxwell's law of distribution of velocity velocities.

The number of molecules lies within box of volume $du dv dw$ whose velocity lies within C to $C+dC$ is the same as number of molecules within the spherical shell of inner radius C and outer radius $C+dC$. Hence volume $du dv dw$ can be replaced volume of the shell $4\pi C^2 dC$. Therefore eqn (v) becomes.

$$dN = N \left(\sqrt{\frac{m}{2\pi kT}} \right)^3 e^{-\frac{mC^2}{2kT}} 4\pi C^2 dC$$

$$dN = 4\pi N \left(\sqrt{\frac{m}{2\pi kT}} \right)^3 e^{-\frac{mC^2}{2kT}} C^2 dC$$

The above equation is called as Maxwell's equation of distribution of velocity which gives the value of number molecules dN having velocity C to $C+dC$.



Mean or Average Velocity C_{av}

The average speed is the sum of all the velocities ranging from 0 to ∞ divided by the total number of molecules N .

$$\therefore C_{av} = \frac{\int_0^\infty dN C}{N}$$

Substituting the value of dN in above equation.

$$C_{av} = \frac{\int_0^\infty \left(4\pi N \left(\sqrt{\frac{m}{2\pi kT}} \right)^3 e^{-\frac{mC^2}{2kT}} C^2 dC \right) C}{N}$$

$$C_{av} = 4\pi \left(\sqrt{\frac{m}{2\pi kT}} \right)^3 \int_0^\infty e^{-\frac{mC^2}{2kT}} C^3 dC$$

$$= 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \int_0^\infty e^{-\frac{mC^2}{2kT}} C^3 dC \quad (39)$$

$$\int_0^{\infty} e^{-bc^2} c^3 dc = \frac{1}{2b^2} \quad (\text{where } b = \frac{m}{2kT})$$

$$C_{av} = 4\pi \left(\frac{b}{\pi}\right)^{\frac{3}{2}} \frac{1}{2b^2}$$

$$C_{av} = \sqrt{\frac{11}{13}}$$

$$C_{av} = \sqrt{\frac{4}{\pi \left(\frac{m}{2kT}\right)}} = \sqrt{\frac{8kT}{\pi m}}$$

where m is the mass of gas molecule. If M is molecular weight which can be given by $M = m N_A$, then $m = \frac{M}{N_A}$. Therefore the above equation becomes,

$$C_{av} = \sqrt{\frac{8kT}{\pi \left(\frac{M}{N_A}\right)}}$$

$$C_{av} = \sqrt{\frac{8kT}{\pi M}} \quad (K_A = R)$$

Rms mean Square Velocity (C_{rms})

The rms speed is the square root of the sum of all the squared velocities ranging from 0 to ∞ divided by the total number of molecules N .

$$C_{rms} = \sqrt{\frac{\int_0^{\infty} c^2 dN}{N}} \times c^2$$

Substituting the value of dN is above equation

$$C_{rms} = \sqrt{\int_0^{\infty} \left(4\pi N \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{mc^2}{2kT}} c^2 dc\right) \times c^2}$$

$$= \sqrt{4\pi \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \int_0^{\infty} e^{-\frac{mc^2}{2kT}} c^4 dc}$$

$$= \sqrt{4\pi \left(\frac{b}{\pi}\right)^{\frac{3}{2}} \int_0^{\infty} e^{-bc^2} c^4 dc} \quad (\text{where } b = \frac{m}{2kT})$$

$$= \sqrt{4\pi \left(\frac{b}{\pi}\right)^{\frac{3}{2}} \times \frac{3}{8} \left(\frac{\pi}{b^3}\right)^{\frac{1}{2}}}$$

$$= \sqrt{\frac{3}{2b}}$$

$$= \sqrt{\frac{3}{2 \left(\frac{m}{2kT}\right)}}$$

$$C_{rms} = \sqrt{\frac{3kT}{m}}$$

Substituting $m = \frac{M}{N_A}$ in above equation we get

$$C_{rms} = \sqrt{\frac{3kT}{\left(\frac{M}{N_A}\right)}} = \sqrt{\frac{3k N_A T}{M}} = \sqrt{\frac{3RT}{M}} \quad (K_A = R)$$

Most probable velocity (C_{pr})

The most probable velocity is the velocity possessed by maximum number of molecules. For this condition $\frac{dN}{dc} = 0$

$$\frac{dN}{dc} = 0$$

$$\frac{d}{dc} \left(4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} e^{-\frac{mc^2}{2kT}} c^2 dc \right) = 0$$

$$4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \frac{d}{dc} \left(e^{-\frac{mc^2}{2kT}} c^2 dc \right) = 0$$

$$4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \left(e^{-\frac{mc^2}{2kT}} [2c] + \left[-\frac{2mc}{2kT} \right] e^{-\frac{mc^2}{2kT}} c^2 \right) = 0$$

$$4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} e^{-\frac{mc^2}{2kT}} \left(2c - \left[\frac{mc}{kT} \right] c^2 \right) = 0$$

$$2c - \left[\frac{mc}{kT} \right] c^2 = 0$$

$$\therefore \frac{mc^3}{kT} = 2c$$

$$\therefore c^2 = \frac{2kT}{m}$$

$$\therefore c = \sqrt{\frac{2kT}{m}}$$

The most probable velocity,

$$C_{pr} = \sqrt{\frac{2kT}{m}}$$

Substituting $m = \frac{M}{N_A}$ in the above equation. We get.

$$C_{pr} = \sqrt{\frac{2kT}{\left(\frac{M}{N_A}\right)}} = \sqrt{\frac{2kN_A T}{M}} = \sqrt{\frac{2RT}{M}} \quad (N_A = R)$$

Relation between C_{pr} , C_{av} , and C_{rms} .

$$C_{pr} : C_{av} : C_{rms} = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{M}} : \sqrt{\frac{3RT}{M}}$$

$$= \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$$

$$= 1 : \frac{2}{\sqrt{\pi}} : \sqrt{\frac{3}{2}}$$

$$= 1 : 1.128 : 1.225$$

Brownian Motion

Robertson Brown observed under a high power microscope that when pollen grains suspended in liquid were moving irregular and constantly in random fashion in all directions. This irregular motion of particles suspended in liquid is called Brownian Motion.

Salient features of Brownian motion

- (i) The motion is Continuous, irregular and random.
- (ii) The motion is independent of mechanical vibration.
- (iii) The smaller the size of the particle, greater is the motion and vice-versa.
- (iv) The lower is the viscosity of the liquid, greater is the motion and vice-versa.
- (v) The higher is the temperature, the greater is the motion and vice-versa.

Important of Brownian Motion

- (i) It is evidence in the existence of molecules which are too small to be observed directly.
- (ii) It is evidence that molecules are not still, but are in Continuous motion.

Real Gases

- (i) These gases which do not follow ideal gas equations under all conditions of temperature and pressure are called real gases.

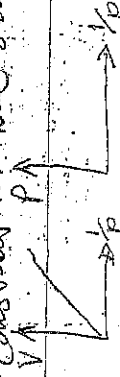
(ii) Real gases exist in nature.

- (iii) The volume occupied by gas molecules is not negligible in comparison to the total volume of container.

- (iv) The force of attraction between real gas molecules is not negligible.

Boyle's Law

Boyle's law states that the volume of a given mass of gas is inversely proportional to its pressure, provided the temperature remains constant or the same.



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Ideal Gases

These gases which follow ideal gas laws and equation under all conditions of temperature and pressure are called real ideal gases.

There is no existence of ideal gases.

The volume occupied by ideal gas molecules is negligible in comparison to the total volume of container.

The force of attraction between ideal gas molecules is negligible.



When the volume of a gas is reduced or decreased, the molecules take less time to travel between the piston and the wall opposite - Consequently, the impacts per second are made. Thus, the pressure is increased, when the volume of the gas is increased, the molecules take a longer time to move from one to the piston and there are fewer impacts per second. Thus the pressure of the gas is reduced or decreased.

Question 4: Find mass of gas occupies a volume of 20 cm³ at a pressure of 750 mmHg. Assume that the temperature remains constant. What will be the volume of the gas at 750 mmHg?

$$P_1 V_1 = P_2 V_2$$

$$750 \times 20 = 750 \times V_2$$

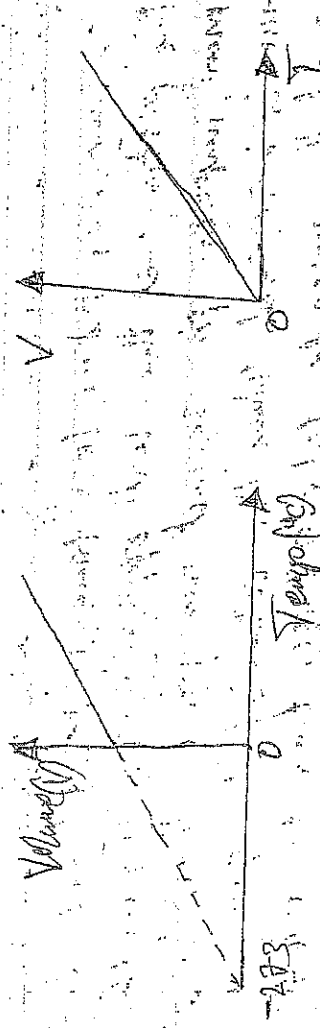
$$V_2 = 20 \text{ cm}^3$$

Charles' Law states that the volume of a gas varies directly as the absolute temperature, provided the pressure is constant. Thus, the volume of a gas is directly proportional to its absolute temperature.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{20}{273} = \frac{V_2}{273}$$

$$V_2 = 20 \text{ cm}^3$$



In terms of kinetic theory, when gases are heated at constant pressure, the molecules gain kinetic energy (KE) and move faster, so the average momentum change of the molecules on impact with the walls is the same as before.

Thus, the volume increases so that the molecules have to travel further between collisions with the walls for successive impacts.

Absolute zero

With n and p constant (Charles's law), the volume of an ideal gas decreases linearly with T and if the gas remained ideal, would reach zero at $T = 0K$. Similarly, with n and V constant

(Gay-Lussac's law), the pressure would decrease to zero with the temperature. This infinite temperature at which p and V would reach zero is called absolute zero.

Question The volume of a fixed mass of gas is $100cm^3$ at $27^\circ C$. Calculate its temperature when the volume is reduced to $\frac{1}{2}$ the original volume of the same pressure. Solve

$$V_1 = 205cm^3, T_1 = 273 + 27 = 300K, T_2 = ?, V_2 = \frac{1}{2} \times 205cm^3$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \frac{205}{300} = \frac{102.5}{T_2} \quad T_2 = \frac{300 \times 102.5}{205} = 150K$$

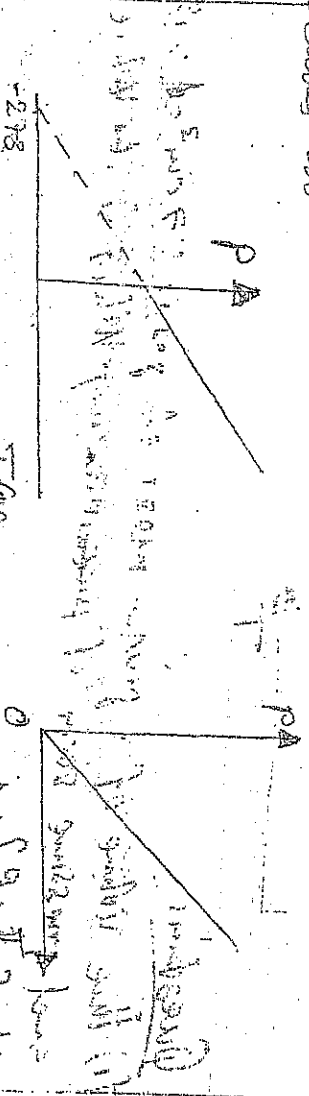
Pressure Law

Pressure law states that the pressure of a gas at constant volume is directly proportional to its absolute temperature T .

Hence, $P \propto T$ (Volume Constant)

$$\frac{P}{T} = k \Rightarrow \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

In terms of kinetic theory of gases, when gases are heated, the molecules gain kinetic energy (K.E) and move faster, so that the momentum change produced by the molecules at the wall per impact is greater. Thus, the number of impacts per second of each molecule also increases since the speed increases.



Question: A fixed mass of gas at STP is heated such that its volume remains unchanged. Calculate the temperature when its pressure is increased to 1.14 atm of pressure.

Given: (4g) 200g

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{1.013 \text{ atm}}{273 \text{ K}} = \frac{1.14 \text{ atm}}{T_2}$$

$$T_2 = \frac{1.14 \times 273}{1.013} = 300.15 \text{ K}$$

$$T_2 = 300.15 \text{ K} = 27.02^\circ \text{C}$$

$$T_2 = 27.02^\circ \text{C}$$

$$T_2 = 27.02^\circ \text{C}$$

$$T_2 = 27.02^\circ \text{C}$$

$$T_2 = 27.02^\circ \text{C}$$

General Gas Law

$V \propto T$ (Pressure Constant) Charles's Law

$P \propto \frac{1}{V}$ (Volume Constant) - Pressure Law

Combining these gives:

$$PV \propto T$$

$$\frac{PV}{T} = k \Rightarrow P_1 V_1 = P_2 V_2$$



Questions

(1) The volume of a given mass of gas is 57 cm^3 at 20°C and pressure 800 mm of mercury. What is the volume at 12°C and 760 mm of mercury?

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{800 \times 57}{273 + 20} = \frac{760 \times V_2}{273 + 12}$$

$$V_2 = \frac{800 \times 57 \times 273}{760 \times 285}$$

$$V_2 = \frac{800 \times 57 \times 273}{760 \times 285} = 54.8 \text{ cm}^3$$

(2) Air at temperature 52°C and pressure 30 atmosphere is admitted into the cylinder of an engine. Calculate the pressure of the gas when it has expanded to 5 times its volume and cooled to 12°C . What becomes the engine?

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{30 \times V_1}{273 + 52} = \frac{P_2 \times 5V_1}{273 + 12}$$

$$P_2 = \frac{30 \times 273}{5 \times 285} = 5.7 \text{ atm}$$

Heat Current

The heat current is the rate at which heat is transferred over time. Because it is a rate of heat energy over time. The S.I. unit of heat current is Joule per second, or Watt (W).

Heat flows through material object through conduction, like heated particles imparting their energy to neighboring particles. Scientists study the flow of heat through materials well before they even knew that the material were made up of atoms, and heat current is one of the concepts that was helpful in this regard.

Even today, though we understand heat transfer to related to the movement of individual atoms, it is most situations it is impractical and unhelpful to try to think of the situation in the way of a single particle to deal the situation on a larger scale in the most appropriate way to study & predict the movement of heat.

Mathematics of Heat Current

Because heat current represents the flow of heat energy over time, you can think about it as representing a tiny amount of heat energy dQ (J) is the variable commonly used to represent heat energy. Transmitted over a tiny amount of time dt . Using the variable H to represent heat current, this gives you the equation,

$$H = \frac{dQ}{dt}$$

If you're taken pre-calculus or Calculus, you might realize that a rate of change like this is a pretty example of when you would want to take a limit as the time approaches zero. Experimentally, you can do that by measuring the heat change

at smaller and smaller time intervals. Experiments conducted to determine the heat current have identified the following mathematical relationship:

$$H = \frac{dQ}{dt} = KA(T_H - T_C) / L$$

That may seem like an intimidating array of variables, so let's break those down (some of which have already been explained).

H is heat current

dQ is small amount of heat transferred over time dt

dt is small amount of time over which occurs transferred

K is Thermal Conductivity of the material

A is Cross-sectional area of the object

$T_H - T_C$ is the temperature difference between the warmest and coldest temperature in the material

L is the length across which the heat is being transferred

There's one element of the equation that should be considered independently: $(T_H - T_C) / L$

This is the temperature difference per unit length, known as temperature gradient.

Thermal Resistance

In engineering, they often use the concept of thermal resistance, R , to describe how well a thermal insulator prevents heat flow from transferring across the material from a stop

(#8)

of material of thickness L , the relationship for a given material is $R = \frac{L}{k}$ resulting in this relationship.

$$H = A \left(\frac{T_1 - T_2}{L} \right)$$

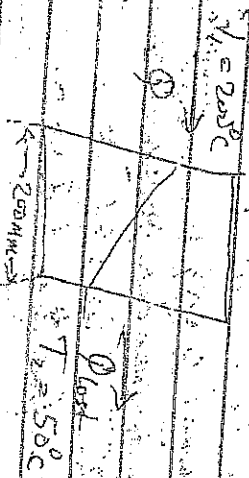
R

where R is the thermal resistance,
 A is the area.

Examples

(1) The inner surface of furnace wall is at 200°C and outer surface at 50°C . Calculate the heat loss per m^2 area of the wall. If the thermal conductibility of the brick is $0.5 \text{ W/m}^\circ\text{C}$, the wall thickness is 200 mm .

85 W

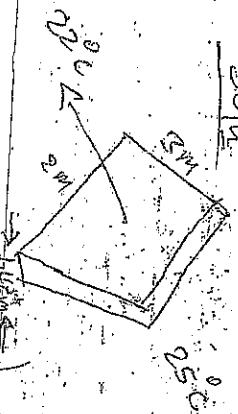


$T_1 = 200^\circ\text{C}$, $T_2 = 50^\circ\text{C}$, $L = 200 \text{ mm}$ (0.2 m), $k = 0.5 \text{ W/m}^\circ\text{C}$

Heat lost $\text{W/m}^2 = \frac{k(T_1 - T_2)}{L} = \frac{0.5(200 - 50)}{0.2} = \underline{\underline{375 \text{ W/m}^2}}$

(2) What is the rate of heat flow through a glass window that is 3 mm thick when the inside temperature is 22°C and the outside temperature is 25°C ?

50 W



(44)

$$\frac{\Delta Q}{\Delta t} = \frac{KA \Delta T}{L} \quad \boxed{W/s = W}$$

$$= \frac{(0.84 W)}{(5 \pi \times 10^{-6})} (6 \pi^2) (30) = 1080 \text{ watt}$$

$$0.014 W$$

If 1080 J of heat energy is transferred through the window every second, how much energy will be transferred in a month?

$$P = \frac{E}{t} \quad E = Pt$$

$$\frac{1 \text{ month} \times 30 \text{ days} \times 24 \text{ hr} \times 3600 \text{ s}}{1 \text{ month} \times 1 \text{ day} \times 1 \text{ hr}} = 2,592,000 \text{ s}$$

$$E = 1080 \times 2,592,000 \text{ sec} = 2.799 \times 10^9$$

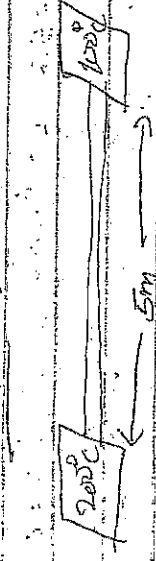
(3) * 5m long Circular Copper rod and inside of Copper has a radius of 5cm. The rod is placed between 2 metal at 200°C and another metal at 100°C. The thermal conductivity of the Copper is 385 W/m°C.

(a) Calculate the heat current in the rod.

(b) How much thermal energy will be transferred in 5 min.

(c) Calculate the temperature gradient.

(d) What will be the steady state temperature in the Copper rod 2m from the hot metal?



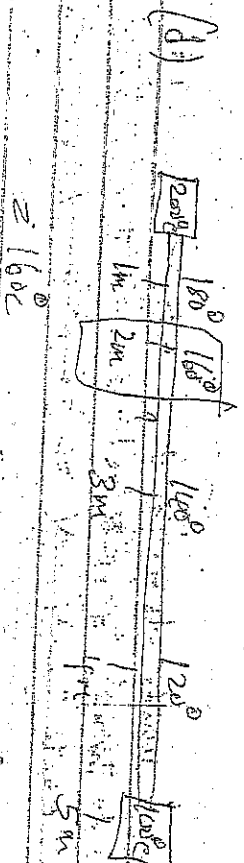
$$(a) H = \frac{\Delta Q}{\Delta t} = \frac{KA(T_H - T_C)}{L}$$

$$= (385) \left[\frac{\pi (0.05 \text{ m})^2}{5 \text{ m}} \right] (200^\circ\text{C} - 100^\circ\text{C}) = 5744.13 \text{ W}$$

$$(b) Q = Ht$$

$$Q = 5 \text{ mm} \times 60 \text{ s} \times 5744.13 \text{ W} = 163.290 \text{ J}$$

$$(c) \frac{\Delta T}{L} = \frac{100^\circ\text{C}}{5 \text{ m}} = 20^\circ\text{C/m}$$



Thermal Conductivity of a material

Thermal Conductivity refers to the ability of a given material to conduct transfer heat. It is generally denoted by the symbol " k " but can also be denoted by " λ " and " κ ". The reciprocal of this quantity is known as thermal resistivity. Material with high thermal conductivity are used as heat sinks where as material with low value of k are used as thermal insulators. Fourier's law of thermal conductivity (also known as the law of heat conduction) states that the rate at which heat is transferred through a material is proportional to the negative of the temperature gradient and is inversely proportional to the area through which the heat flows. The above formula of this law can be expressed through the following equation;

$$Q = -k \frac{A}{L} \Delta T$$

Where ΔT refers to the temperature gradient.

Q denotes the thermal flux or heat flow and k refers to the thermal conductivity of the material.

A = area through which the heat flows
 L = length

Thermal Conductivity is expressed in terms of the following dimensions;

- Temperature, Length, mass and Time
- The S.I unit of this quantity is watt per meter-kelvin ($Wm^{-1}K^{-1}$)
- It is generally expressed in term of power / (length temperature)

These units describe the rate of conduction of heat through a material of unit thickness and for each Kelvin of temperature difference.

Every substance has its own capacity to conduct heat, the thermal conductivity of a material is described by the following formula

$$K = \frac{QL}{A \Delta T}$$

K is the thermal conductivity in W/mK

Q = thermal amount of heat transferred through the material in Joules second or Watt

L = distance between the two isothermal planes

A = area of the surface in square meter

ΔT is the difference in temperature between measurement

There exist several methods of measuring the thermal conductivity of materials. These methods are broadly classified into two types of techniques.

(i) Transient and

(ii) Steady state techniques

Factors affecting thermal conductivity

(i) Temperature effect: The thermal conductivity of metals and non metals will not be the same.

(ii) The chemical phase of the material

(iii) Thermal anisotropy (different in coupling of phonons)

(iv) The electrical conductivity of the material.

(v) Influence of magnetic field.

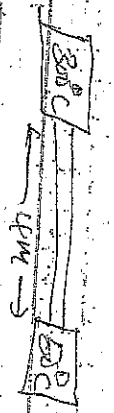
(vi) Isotopic purity of crystal.

Example 4

A 4m long aluminium rod with a cross-sectional area of 0.05m^2 attached to a hot metal at 800°C and a cool metal at 50°C .

(3) Calculate the amount of thermal energy that will be transferred in 15 second.

(b) determine thermal resistance of the Aluminum rod.



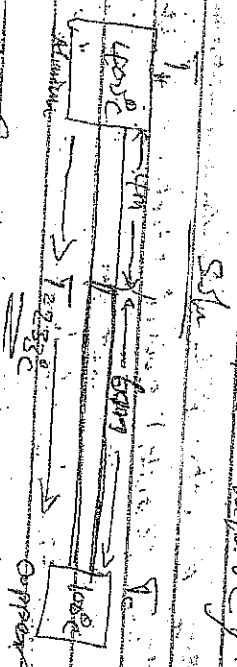
$k = 960 \frac{W}{m \cdot ^\circ C}$
 $\delta = 1.55$
 rod curved

(a) $Q = kA \frac{T_h - T_c}{L}$ $k = 205 \frac{W}{m \cdot ^\circ C}$

$Q = (205)(0.05)(15)(25) = 3960 J$

(b) $R = \frac{L}{kA} = \frac{4m}{205 \cdot 0.0195m^2} = 0.0195m^2/W$

(5) An aluminum bar is attached to a hot reservoir at 400°C. The aluminum bar is attached to a low copper bar which is attached to a reservoir of 100°C. The two metal bars have the same cross-sectional area. What is the steady state temperature of the aluminum copper junction. Thermal conductivities of Aluminum and Copper are 205 & 385 W/m°C.



$kA [T_h - T] = kA [T - T_c]$

$205 [400 - T] = 385 [T - 100]$

$81250 - 205T = 385T - 38500$

$20500 - 5125T = 6417T - 6417$

$26917 = 11542T$

$T = 233^\circ C$

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Watt's emit less light

Stefan's Boltzmann Law
Statement that the total radiant heat power emitted from a surface is proportional to fourth power of its absolute temperature
 $E = \sigma T^4$

σ is constant proportionality called the Stefan's Boltzmann Constant.

$E =$ emissivity (1 for black body & less than 1 for gray bodies).

Watt's Constant has value 5.67×10^{-8}

Watt per meter² per K⁴.

This law applicable as applies only to black bodies theoretical surfaces that absorb all incident heat radiation.

Black body Radiation
Energy radiated by any object or system that absorbs all incident radiation. The term used by us refers to the spectrum of light emitted by any heated object. Common examples include the heating element of a toaster and filament of a light bulb. The Spectra of incandescent black body radiation peaks at a temperature that the emitting body. Room temperature objects (about 300K) emit radiation with a peak intensity in the far infrared. Radiation from toaster filaments and light bulb filaments about 2000K and 2700K respectively also peak in the infrared, though their spectra extend progressively into the visible; the 6000K surface of the sun emits blackbody radiation that peaks in the visible range.

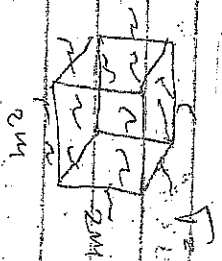
Factors that effect the amount of radiation of particular wavelength given off from a surface include the type of material absorbing the incident radiation and the temperature of the surface.

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It means that difference amounts of energy between
 If the materials are at the same temperature.
 An object that gives off the maximum amount
 of energy for its absolute temperature is termed
 a blackbody.

Example (6)

What is the total energy emitted by the
 following cube in 1 minute?



Soln

We assume the cube is a perfect blackbody.

$$T = 300^\circ\text{C} + 273.15\text{K} = 573.15\text{K}$$

$$T_2 = 0^\circ\text{C} + 273.15\text{K} = 273.15\text{K}$$

$$P_{\text{em}} E_b = \sigma T^4 = \Phi_{\text{em}} - \Phi_{\text{abs}}$$

$$\therefore \Phi_{\text{em}} = \sigma T^4$$

$$\Phi_{\text{em}} = 0.1 \text{ W}$$

$$\Phi_{\text{em}} = (5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}) (573.15\text{K})^4 (6.0\text{s})$$

$$= 8810.87 \text{ J}$$

(3) A sphere ($\epsilon = 0.92$) of radius R_{em} is at 22°C .
 Its speed is a room temperature of 22°C .

(b) What is the rate of heat loss leaving the sphere?

(c) What is the rate of heat energy entering the sphere?

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$$k = 0.011$$

(c) What is the net rate of heat energy for the sphere?
 (d) How long will it take for the sphere to lose all its
 joules of energy?

Sketch



$$(a) -\Delta Q = (0.11)^3$$

$$\frac{\Delta Q}{\Delta t} = 0.42 \times (5.67 \times 10^{-8}) (0.785) (2.7 + 2.73)$$

$$= -151.4 \text{ W}$$

$$(b) 0.42 (5.67 \times 10^{-8}) (0.785) (2.75)^4$$

$$= 106.9 \text{ W}$$

$$(c) P = -151.4 \text{ W}$$

$$+ 106.9 \text{ W}$$

$$= -44.5 \text{ W}$$

(d) Use by sphere 44.5 J/s

$$P = \frac{E}{t} \quad E = Pt$$

$$t = \frac{E}{P} = \frac{1 \times 10^6 \text{ J}}{44.5} = 22472 \text{ sec}$$

$$= 22472 \text{ sec} \times \frac{1 \text{ hour}}{3600 \text{ s}} = 6.2 \text{ hours}$$

Greenhouse effect
 is a process that occurs when gases in Earth's atmosphere trap the Sun's heat. This process makes Earth much warmer than it would be without an atmosphere. The greenhouse effect is one of the things that makes Earth's comfortable place to live.

Greenhouse effect is the mechanism by which thermal radiation from Earth's surface is reabsorbed by greenhouse gases and redirected in all directions. Some of the major greenhouse gases are water vapor, carbon dioxide, methane, and

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Ozone. Most of the solar energy to pass through the earth's atmosphere thereby warming up the earth's surface.

On the other hand, a proportion of the outgoing energy is absorbed at these gases, which is then redirected to earth's surface, therefore further warming up the atmosphere. This is called a Greenhouse effect.

For ages there has been a debate between how much solar energy is redirected to earth and how much is reflected back to the space. With the increased burning of fossil fuel, the percentage of energy absorbable in the atmosphere has significantly increased thereby adding to the greenhouse effect. These greenhouse buildings are made mostly of transparent material that offers plants in the building maximum access to sunlight. When the sunlight touches the solid surfaces inside the greenhouse, the surface absorbs a lot of the energy and converts it into heat (infrared energy).

There are several benefits of the greenhouse effect. This greenhouse advantage, we are growing various crops in an artificial atmosphere. Also, the solar power water heaters are utilizing the greenhouse effect to warm up water, saving 20-30% of energy bill at homes.

Cases that influence the greenhouse effect

- (i) Carbon dioxide (CO_2)
- (ii) Methane (CH_4)
- (iii) Nitrous oxide (N_2O)
- (iv) Ozone (O_3)
- (v) Water vapour (H_2O)

Benefits of greenhouse effect

(1) The greenhouse effect helps to maintain a certain temperature level on Earth's surface, making it habitable for the living beings (human, animal, and plants and other organisms).

(2) The greenhouse effect helps block the harmful solar radiation from reaching planet's surface, where it goes into the water & filter and bounce back most of the unwanted and damaging energy into space.

(3) Ozone, which is one of the crucial greenhouse gases, absorbs the harmful ultra-violet (UV) rays of sun. If there was not any ozone layer in our atmosphere, it would affect the life on Earth.

(4) It has allowed the planet to maintain a moderate temperature on the surface. Thanks to the moderate temperature, the planet's ice has not been melted completely and the polar ice caps are restricted to the polar regions of the planet.

(5) Same as Earth, changes in the climate due to the greenhouse effect, which will disturb the global warming & the increasing amount of greenhouse gases are causing a gradual rise in the Earth's temperature, which is resulting in the organisms started melting and resulting in massive climate changes.

(6) Increased level of CO_2 in the atmosphere of CO_2 , which is one of the common greenhouse gases, is not only affecting the marine trees but also disrupting the photosynthesis process of plants.

(7) Rising water levels, because of the greenhouse effect and global warming, the water levels are rising, putting millions of trees in danger.